The Power of Translation

Although Figure 13.1 schematically conveys the essential content of Witten's discovery, expressed in this way it might strike you like a bit of inside baseball. Before Witten's breakthrough, researchers thought there were five separate versions of string theory; after his breakthrough, they didn't. But if you'd never known that there were five purportedly distinct string theories, why should you care that the cleverest of all string theorists showed they aren't distinct after all? Why, in other words, was Witten's discovery revolutionary as opposed to a modest insight correcting a previous misconception?

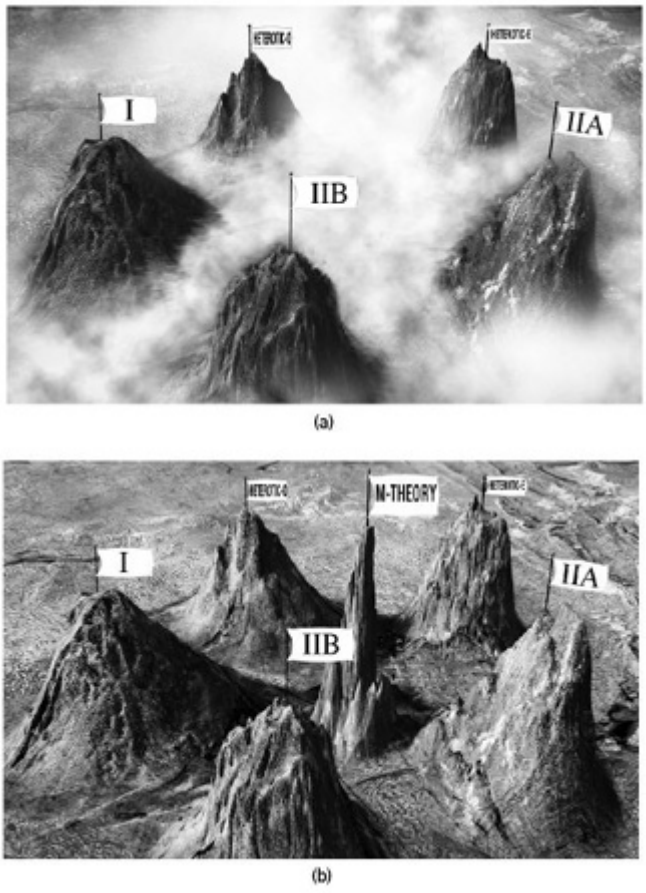


Figure 13.1: **(**a**)** Schematic portrayal of the five string theories, prior to 1995. **(**b**)** Schematic portrayal of the meta-unification revealed by M-theory.

Here's why. Over the past few decades, string theorists have been stymied repeatedly by a mathematical problem. Because the exact equations describing any one of the five string theories have proven so